



Ramrachai Marma

+49 1778365108 | marmaramrachai20@gmail.com | LinkedIn | GitHub | Portfolio | Erlangen, Germany
Work Permit: Germany

PROFESSIONAL EXPERIENCE

Working Student, Data Science

June 2024 – Present

Siemens Mobility GmbH | Erlangen, Germany

- **Data Analytics & Dashboarding:** Analyzed operational and incident datasets using Python and SQL to monitor engineering KPIs, delivering actionable insights via Power BI and Qlik Sense dashboards.
- **ML & LLM Engineering:** Developed and evaluated ML, NLP, and LLM models for information extraction and text classification on structured and unstructured engineering logs.
- **Workflow Automation & CI/CD:** Engineered automated Python data-processing pipelines and integrated them into Azure DevOps CI/CD workflows to improve operational efficiency and data reliability.

Master's Thesis: AI-Assisted Locomotive Fault Reporting System

January 2025 – July 2026

Siemens Mobility GmbH | Erlangen, Germany

- **End-to-End AI Application:** Built an AI-assisted fault reporting system from scratch to convert unstructured German locomotive maintenance logs into structured and validated records using LLM-based extraction and categorization.
- **AI & Data Engineering:** Designed few-shot LLM workflows and an automated document ingestion pipeline to extract information from PDF, DOCX, and JSON sources and map free-text reports to a controlled bilingual engineering taxonomy.
- **Cloud Deployment & Human Validation:** Developed the serverless backend and web interface with Azure OpenAI, Azure Functions, Blob Storage, Azure Document Intelligence, and SharePoint, including a Human-in-the-Loop workflow for expert review and validation.

Data Analyst

May 2021 – May 2022

aiQuest Intelligence | Remote

- **Data Analytics:** Analyzed operational KPIs and learner performance data using quantitative analysis and data validation to support performance monitoring.
- **Data Preparation:** Developed data cleaning and transformation workflows to produce structured analytical reports for management decision-making.

Finance Data Analyst Intern

August 2017 – December 2017

Oasis Group | India

- **Financial Data Analysis:** Conducted financial data audits and discrepancy analysis to identify irregularities, verify reporting consistency, and maintain data integrity.

EDUCATION

M.Sc. Data Science, Major in Artificial Intelligence & Machine Learning

April 2023 – July 2026

Friedrich-Alexander-Universität Erlangen-Nürnberg, Germany

B.Eng. Information Technology

June 2016 – September 2020

Gujarat Technological University, India

PROJECTS

Enterprise RAG & Knowledge Retrieval Pipeline

- **RAG & Knowledge Retrieval:** Built a production-ready RAG system using semantic vector search, document chunking, and secure access governance for high-precision technical document retrieval.

Requirements Specification Extraction Tool

- **Document Intelligence & LLMs:** Developed an automated document-processing tool using Azure Document Intelligence and LLMs to extract, normalize, and deduplicate engineering specifications from multi-format files.

Automated Data Engineering & LLMops Pipeline

- **Data Engineering & LLMops:** Implemented an end-to-end Python data pipeline with CI/CD integration for automated and reproducible data transformation, processing, and schema validation.

Medical Chatbot (Llama 2 RAG)

- **Conversational AI & RAG:** Built a retrieval-augmented Q&A application using a locally hosted Llama 2 model via Hugging Face Transformers, Pinecone, and LangChain, served through a Flask web application.

SKILLS

Machine Learning & GenAI: Large Language Models (LLMs), Azure OpenAI, Azure AI Foundry, Azure Document Intelligence, RAG, LangChain, LlamaIndex, embeddings & vector databases, semantic search, agentic workflows & tool-calling, Model Context Protocol (MCP) & MCP server development, NLP, structured outputs (Pydantic / JSONL), PyTorch

AI Methods & Evaluation: Few-shot prompt engineering, LLM evaluation, human-in-the-loop (HITL) validation, layout-aware document parsing, taxonomy mapping, Agile/Scrum

Programming & Automation: Python, R, VBA, SQL, REST APIs, Streamlit, Power Automate, n8n, GitHub Actions, Azure DevOps, CI/CD

Cloud & Infrastructure: Microsoft Azure, Azure Functions, AWS, Docker, Kubernetes

Data Engineering & Analytics: ETL pipelines, data transformation, quantitative data analysis, MySQL, Snowflake, Power BI, Qlik Sense

Security & Governance: IT security engineering, cloud security, access governance, IT audit, risk assessment, gap analysis

CERTIFICATES

- AWS Cloud Solution Architect Professional Certificate [!\[\]\(97faa0168e491544be255cfcab218e9b_img.jpg\)](#)
- Master Power BI [!\[\]\(b2166b76608b8499cffc130bf1b1fe60_img.jpg\)](#)
- Building Your First AI Agent with OpenAI [!\[\]\(b29da0f81af7d31816596405aed0e378_img.jpg\)](#)
- SQL, MySQL for Data Analytics and Business Intelligence [!\[\]\(52b4a21f1e75ded8f9710f4114e70d28_img.jpg\)](#)
- Introduction to Linux [!\[\]\(07221912d1bf206beb97cefd77af5f78_img.jpg\)](#)

LANGUAGES

English: C2

German: B1+ (Working knowledge; incl. German-language NLP/technical datasets)

AWARDS

Fully Funded Government Scholarship (ICCR, Government of India)

Competitive, merit-based award for study abroad.